

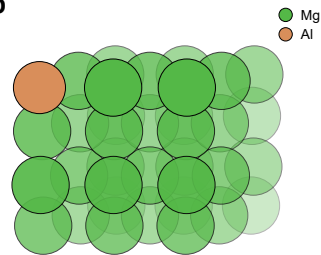
Interface model: Mg(0001)+cation+Al-anion

a



Al in Mg(0001): $\Delta E_{sub} = -4.44$ eV

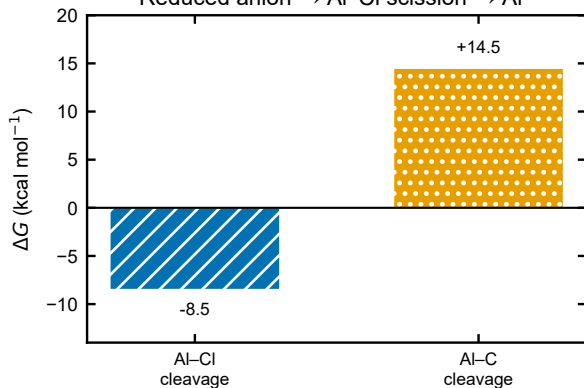
b



● Mg
● Al

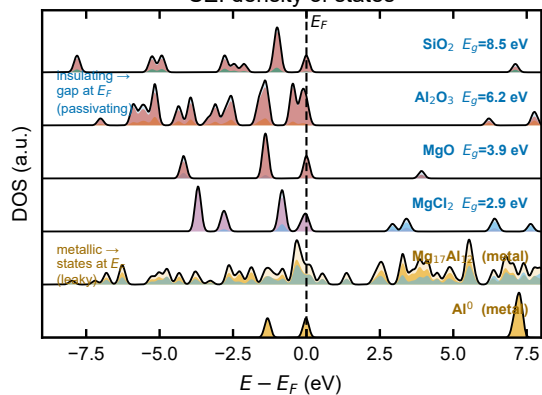
c

Reduced anion $\rightarrow$ Al-Cl scission $\rightarrow$ Al⁰



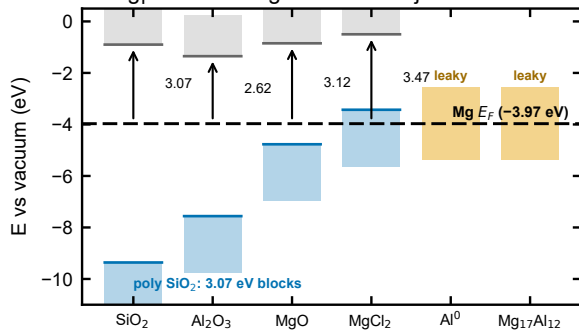
d

SEI density of states



e

Mg|SEI band alignment: e⁻-injection barrier



f

Network inverts the interfacial ion ordering

